Dynamic Semantic Document Segmentation for Academic RAG.

Brådland et al. (2025) introduced the HOPE metric to evaluate how text chunking preserves semantic coherence and topic independence across academic documents. Their empirical benchmarks demonstrate that rigid 256-token fixed-size splitting slices academic arguments across boundaries and lowers downstream answer accuracy by 19%. Wang et al. (2025) resolved boundary slicing by grouping adjacent sentences using cosine similarity thresholds against contextual embeddings.

Duarte et al. (2024) developed LumberChunker to detect narrative and topic shifts dynamically, producing self-contained variable-length passages for dense retrievers. Zhao et al. (2025) further confirmed via Mixtures of Text Chunking Learners that dynamic sentence-boundary segmentation operates with low latency on standard institutional CPU hardware.